Therefore, the ionic product of water at 25°C is

$$K_W=[H_3O]^+[OH^-].....(8.4)$$

$$K_W=(1\times10^{-7})(1\times10^{-7})$$

$$=1\times10^{-14}.$$

Like all equilibrium constants, K_w is also a constant at a particular temperature. The dissociation of water is an endothermic reaction. With the increase in temperature, the concentration of H_3O^+ and OH^- also increases, and hence the ionic product also increases.

K_w values at different temperatures are given in the following below

Temperature (°C)	K_w
0	1.14×10^{-15}
10	2.95×10^{-15}
25	1.00×10^{-14}
40	2.71×10^{-14}
50	5.30×10^{-14}

In neutral aqueous solution like NaCl solution, the concentration of H_3O^+ is always equal to the concentration of OH^- whereas in case of an aqueous solution of a substance which may behave as an acid (or) a base, the concentration of H_3O^+ will not be equal to $[OH^-]$.

We can understand this by considering the aqueous HCl as an example. In addition to the auto ionisation of water, the following equilibrium due to the dissociation of HCl can also exist.

$$HCl+H_2O \rightleftharpoons H_3O^+ + Cl^-$$

In this case, in addition to the auto ionisation of water, HCl molecules also produces H_3O^+ ion by donating a proton to water and hence $[H_3O^+]>[OH^-]$. It means that the aqueous HCl solution is acidic. Similarly, in basic solution such as aqueous NH_3, NaOH etc….. $[OH^-]>[H_3O^+]$.

Example 8.1

Calculate the concentration of OH^- in a fruit juice which contains 2×10^{-3} M, H_3O^+ ion. Identify the nature of the solution.

Given that $H_3O^+=2\times10^{-3}M$

$$K_w=[H_3O^+][OH^-]$$

$$\therefore [OH^-]=\frac{K_w}{[H_3O^+]}=\frac{1\times10^{-14}}{2\times10^{-3}}=5\times10^{-12}M$$

$$2\times10^{-3} >> 5\times10^{-12}$$

i.e., $[H_3O^+]>>[OH^-]$, hence the juice is acidic in nature

XII U8-Ionic equilibrium.indd 8 16-12-2022 19:33:37